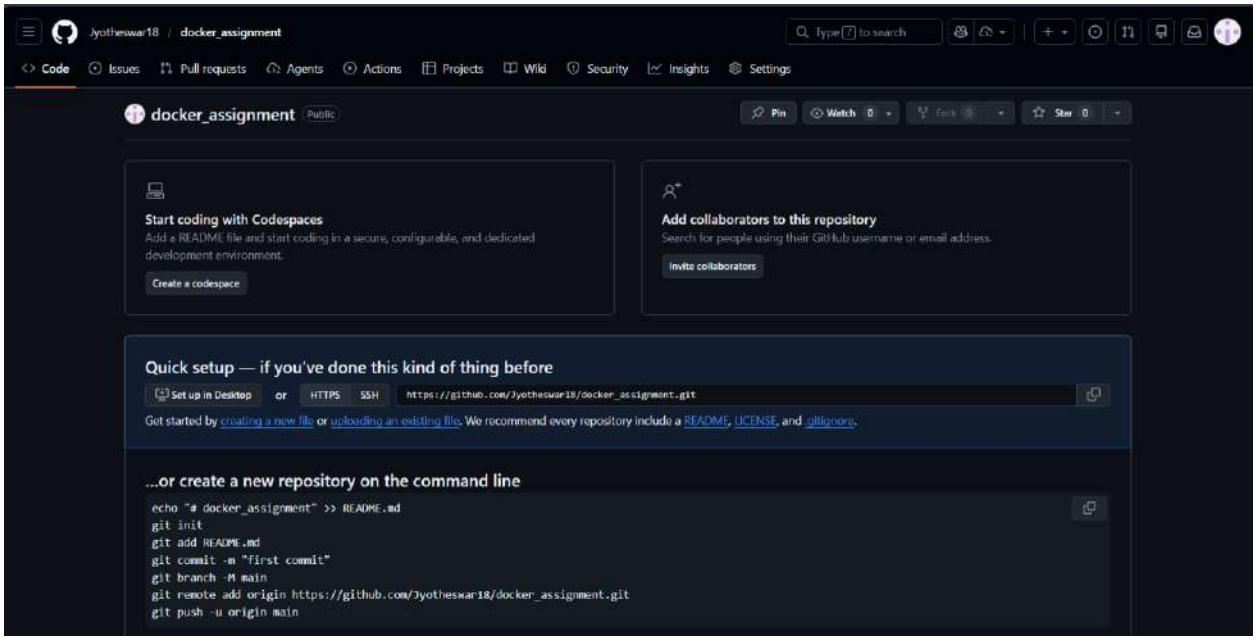
Digital Assignment Lab - 4

Course Name :- Agile Development Process and Devops Lab
Course code :- ISWE406P
Name :- CP. Jyotheswar Reddy
Reg-No:- 23MIS0097

STEP 1: Create GitHub Repository

🔹 **Steps:**

- Go to GitHub
- Click **New Repository**
- Name → docker-example
- Click **Create**



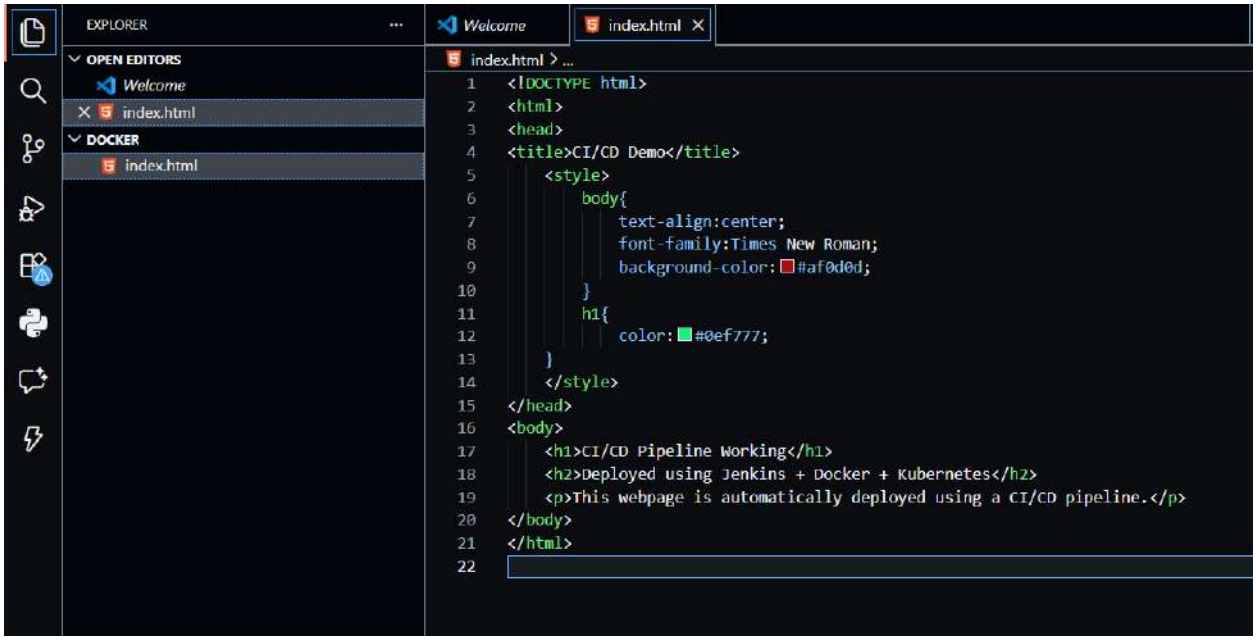
STEP 2: Create Files Locally

- **Create folder:**

docker
Inside create:

1) index.html

```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
   <style>
      body{
         text-align:center;
         font-family:Times New Roman;
         background-color:#af0d0d;
      }
      h1{
         color:#0ef777;
   }
   </style>
</head>
<body>
   <h1>CI/CD Pipeline Working</h1>
   <h2>Deployed using Jenkins + Docker + Kubernetes</h2>
   <p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>
</html>
```



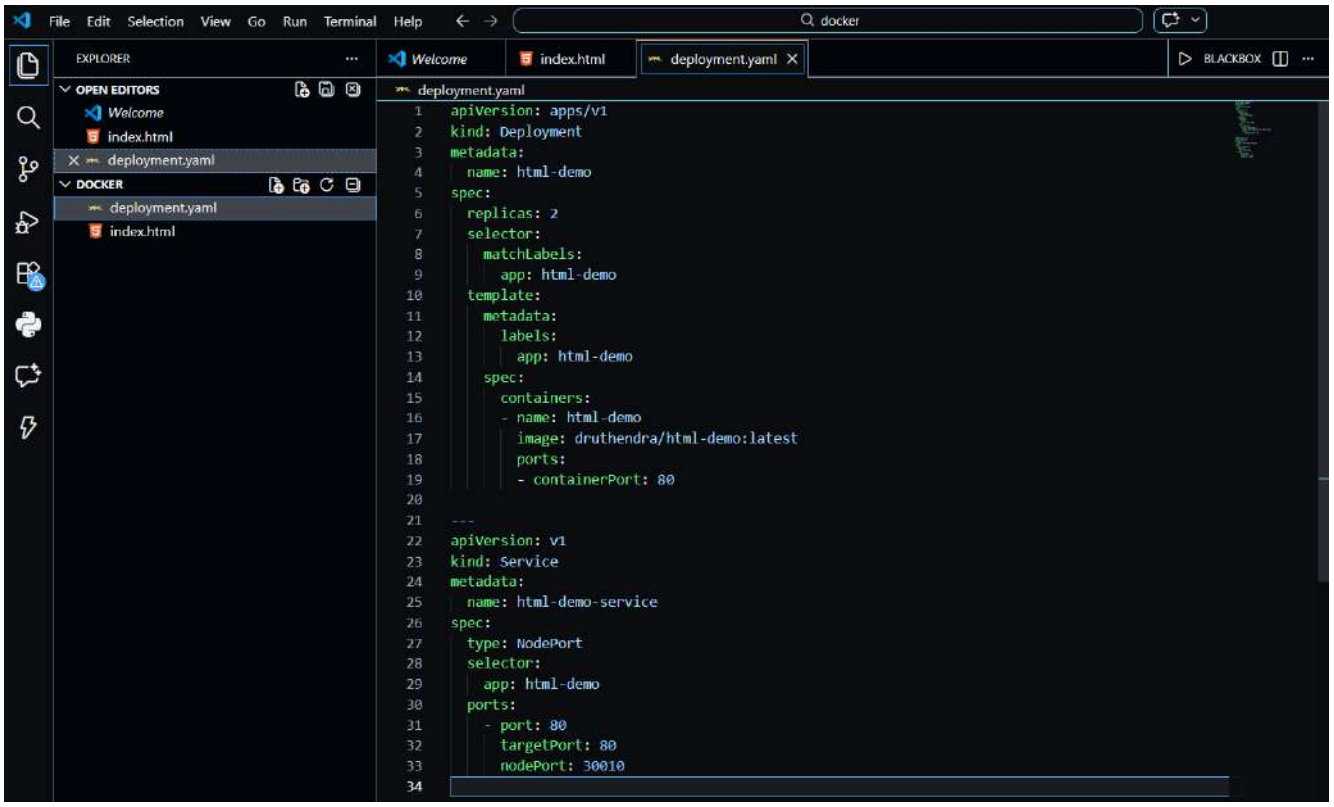
Dockerfile

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
      - name: html-demo
        image: druthendra/html-demo:latest
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
    - port: 80
      targetPort: 80
```

nodePort: 30010



STEP 3: Push Code to GitHub

```
PS E:\docker_assignment> git init
PS E:\docker_assignment> git add .
PS E:\docker_assignment> git commit -m "first commit"
[main (root-commit) 37a26d5f first commit
3 files changed, 58 insertions(+)
create mode 100644 Dockerfile
create mode 100644 deployment.yaml
create mode 100644 index.html
PS E:\docker_assignment> git branch -M main
PS E:\docker_assignment> git remote add origin https://github.com/Jyotheswar18/docker_assignment.git
PS E:\docker_assignment> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Writing objects: 100% (5/5),575), done | 870.0 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Jyotheswar18/docker_assignment.git
*  [new branch]   main -> main
branch 'main' set up to track 'origin/main'.
PS E:\docker_assignment
```

STEP 4: Enable Kubernetes

• Open the Docker Dashboard: Launch the Docker Desktop application on your system.

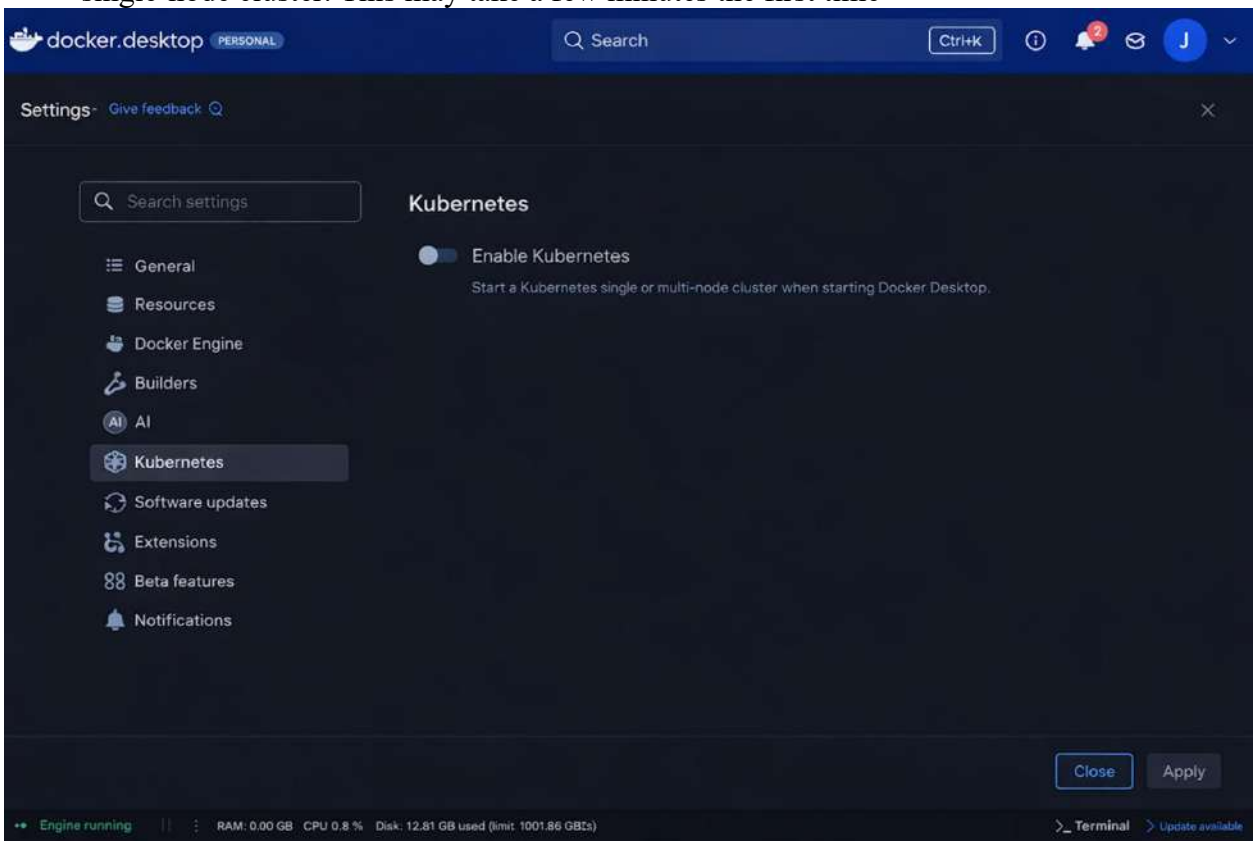
• Navigate to Settings: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.

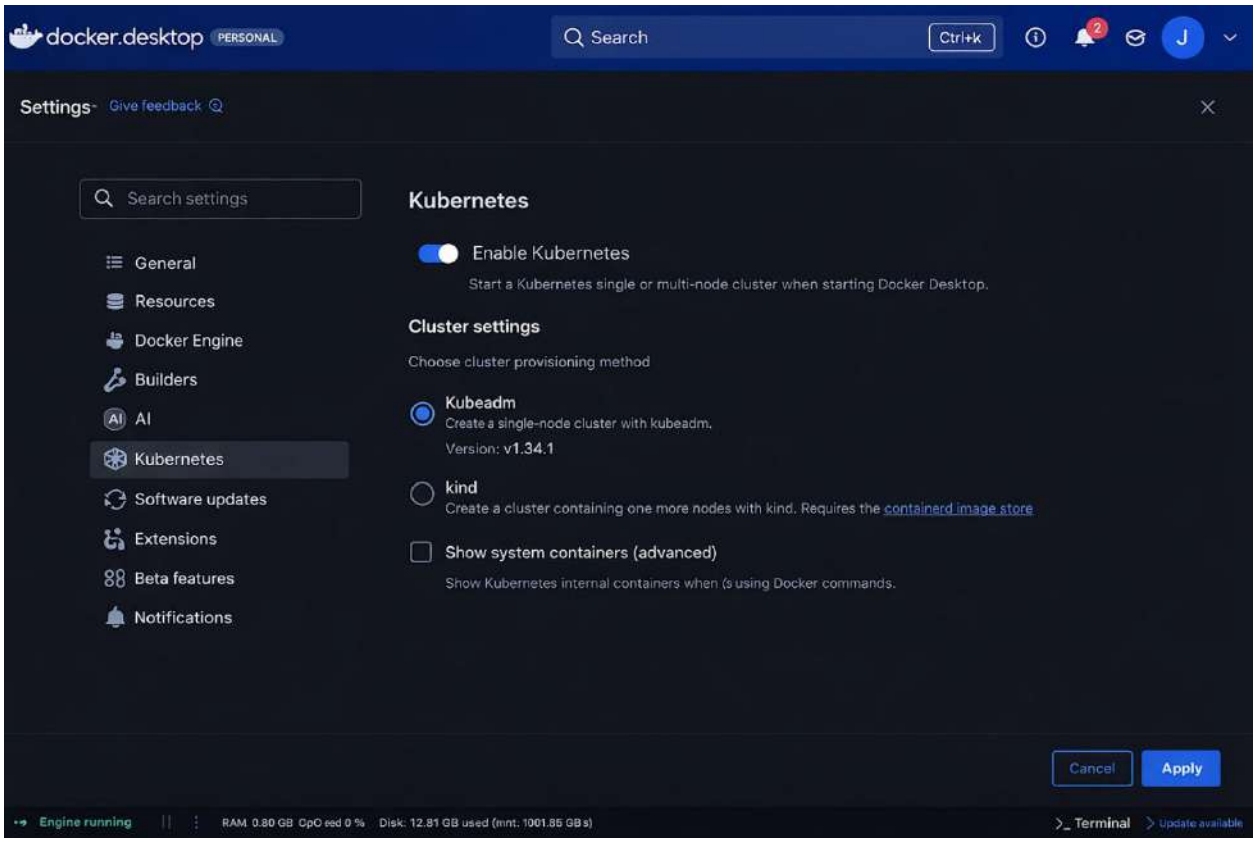
• Select the Kubernetes Tab: In the Settings menu, find and select the Kubernetes tab.

• Enable Kubernetes: Check the box labeled Enable Kubernetes.

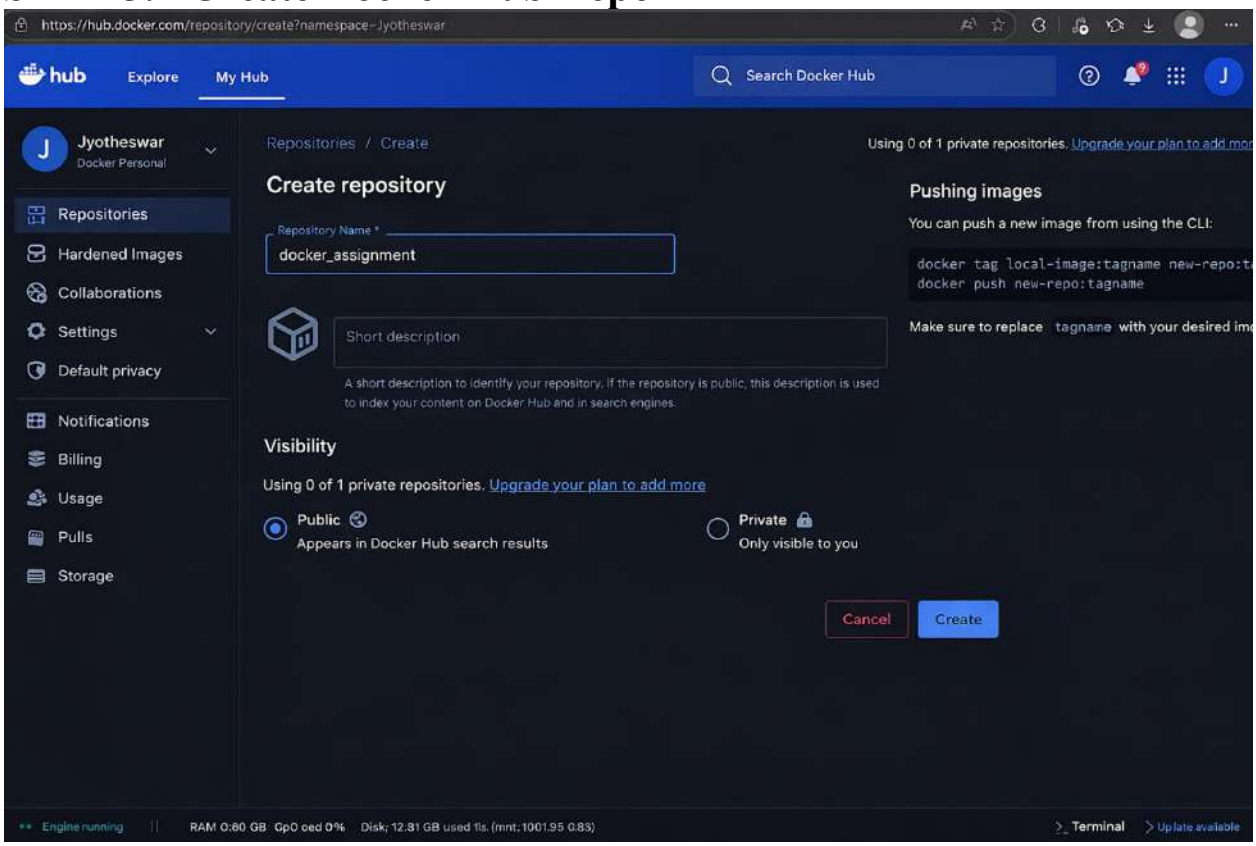
• Apply & Restart: Click the Apply & Restart button at the bottom of the settings window.

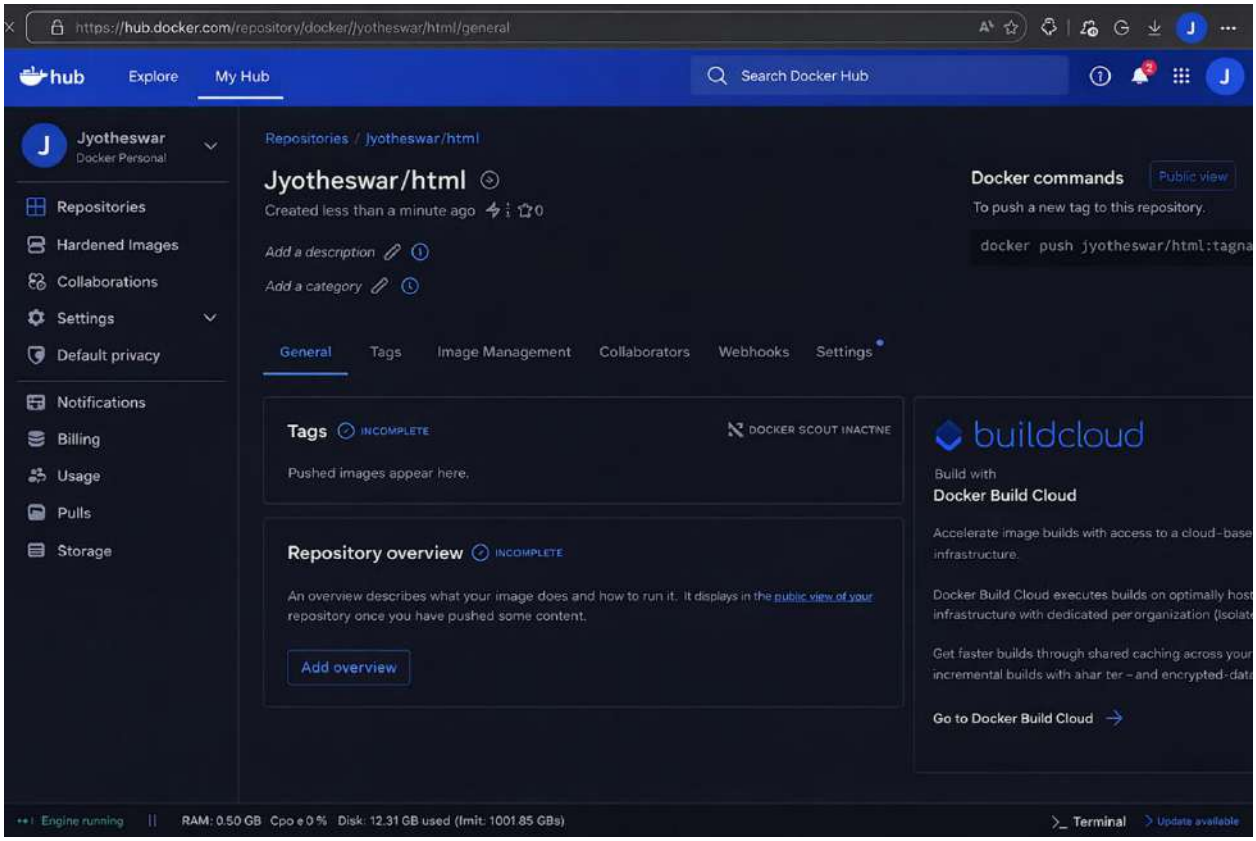
Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time



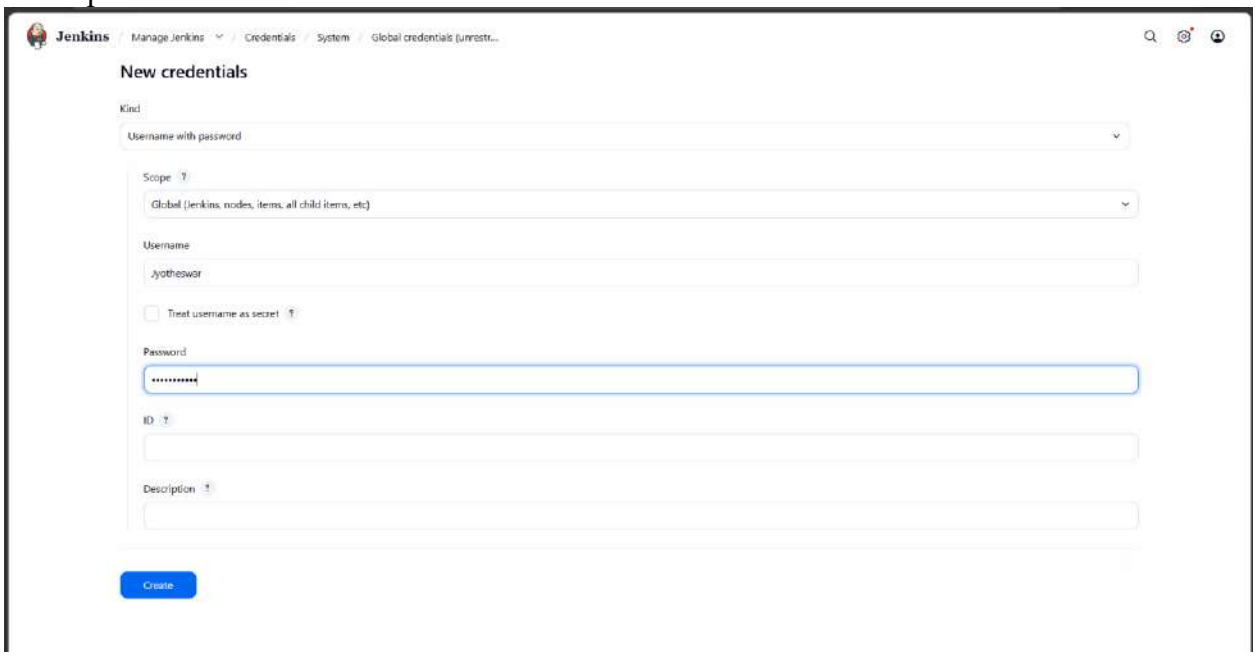


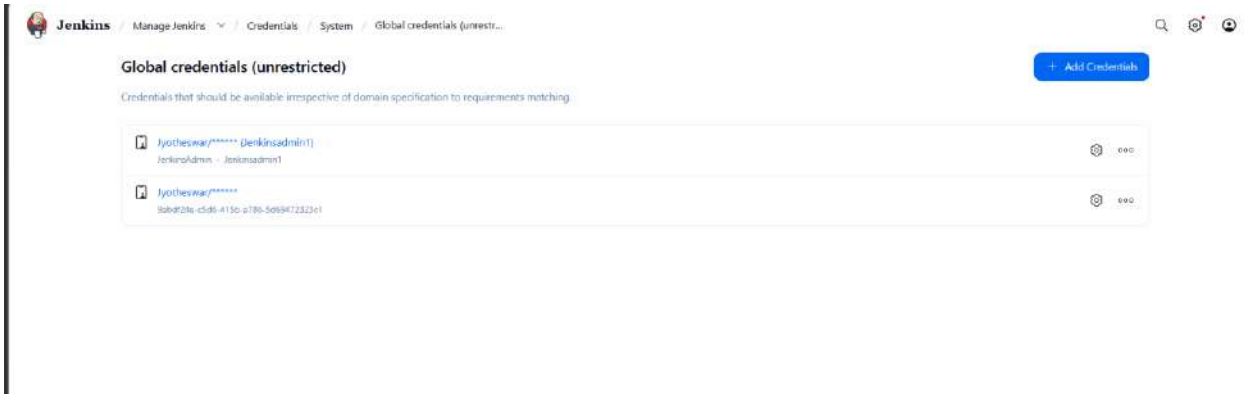
STEP 5: Create Docker Hub Repo



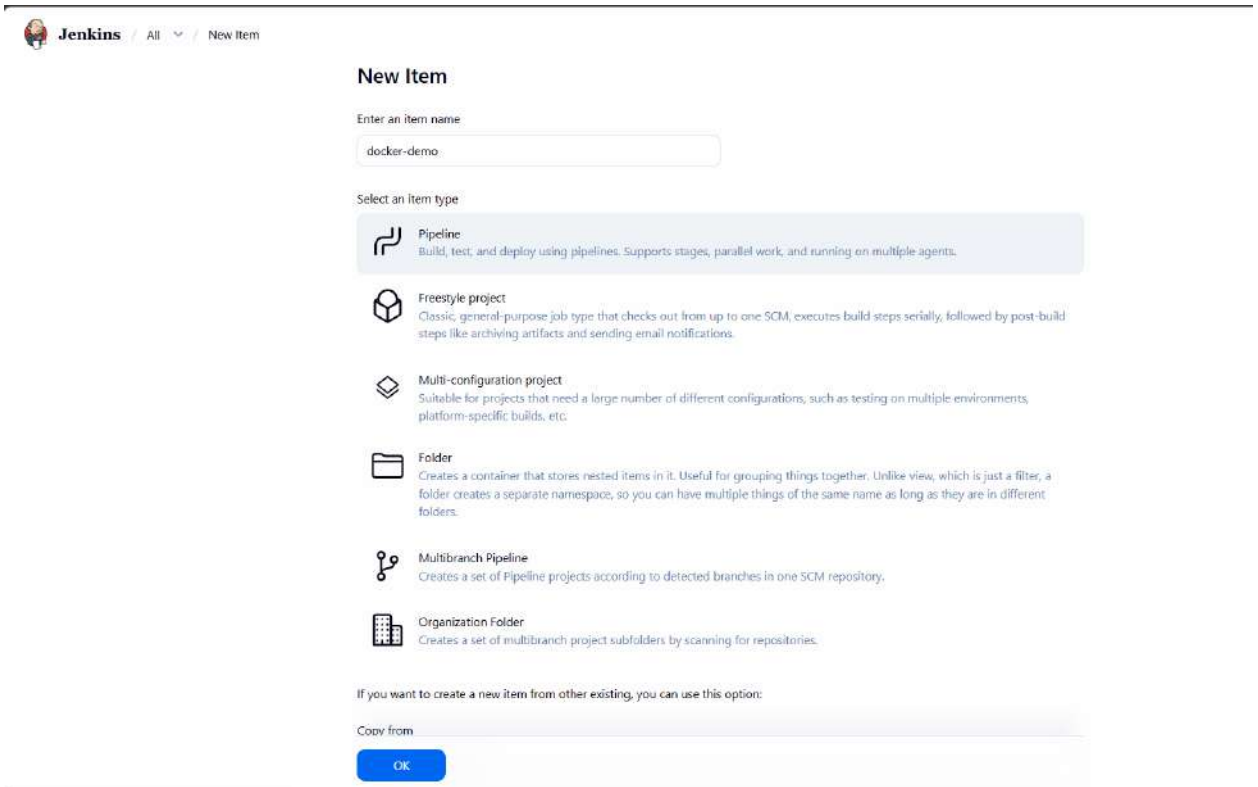


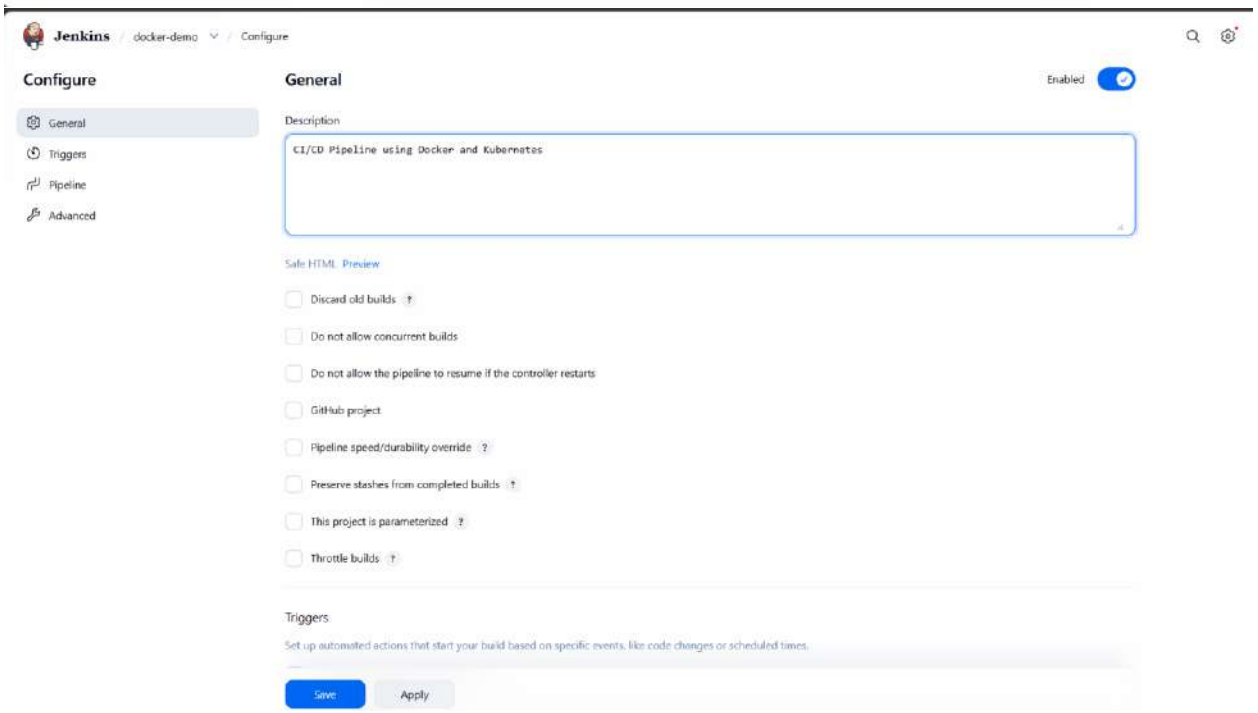
Open Jenkins In Jenkins Setting --> Credentials New Credentials – User name: , password , ID:, Description





Create New Item -> Pipeline Paste the code in pipeline script - - - - Update URL of GIT repository Make sure credentialsId: “dockerhub” same as mentioned in credentials Make sure that the ID mentioned in credentials “kuberconfig” is same as in script Run the Build Pipeline:





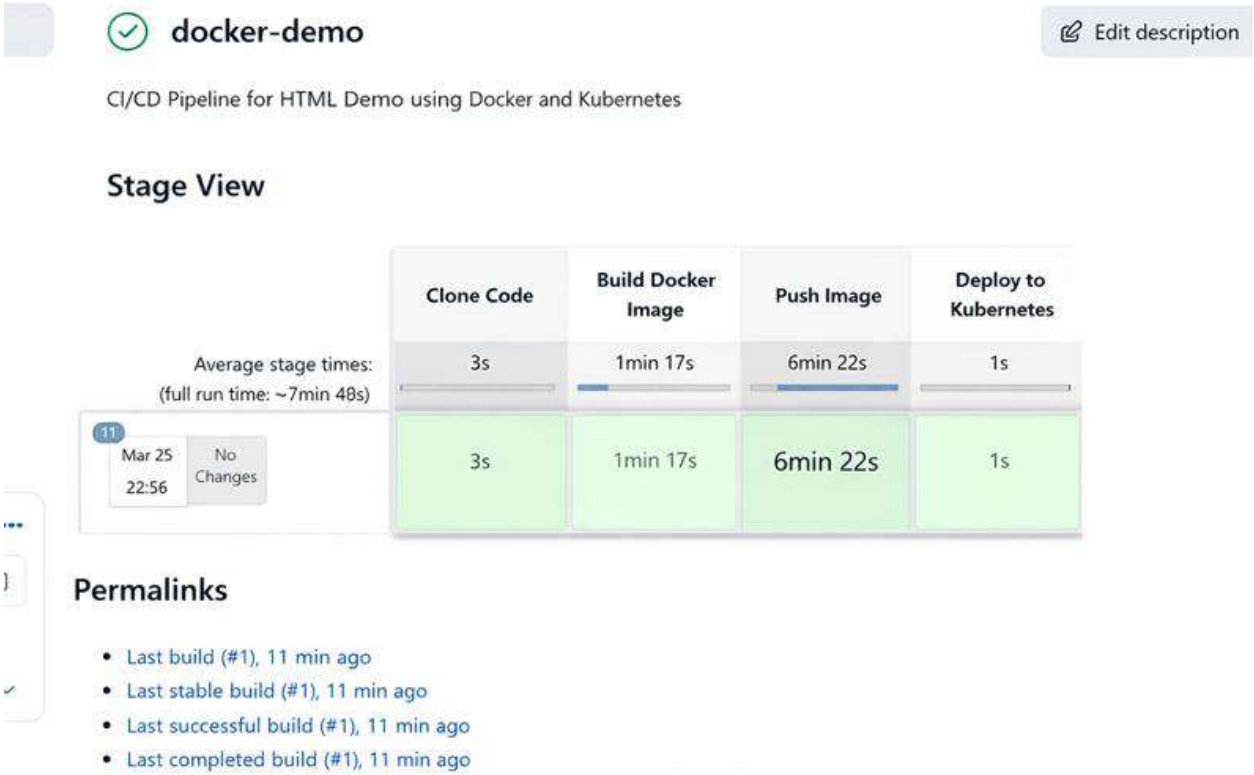
<u>Pipeline Code:-</u>

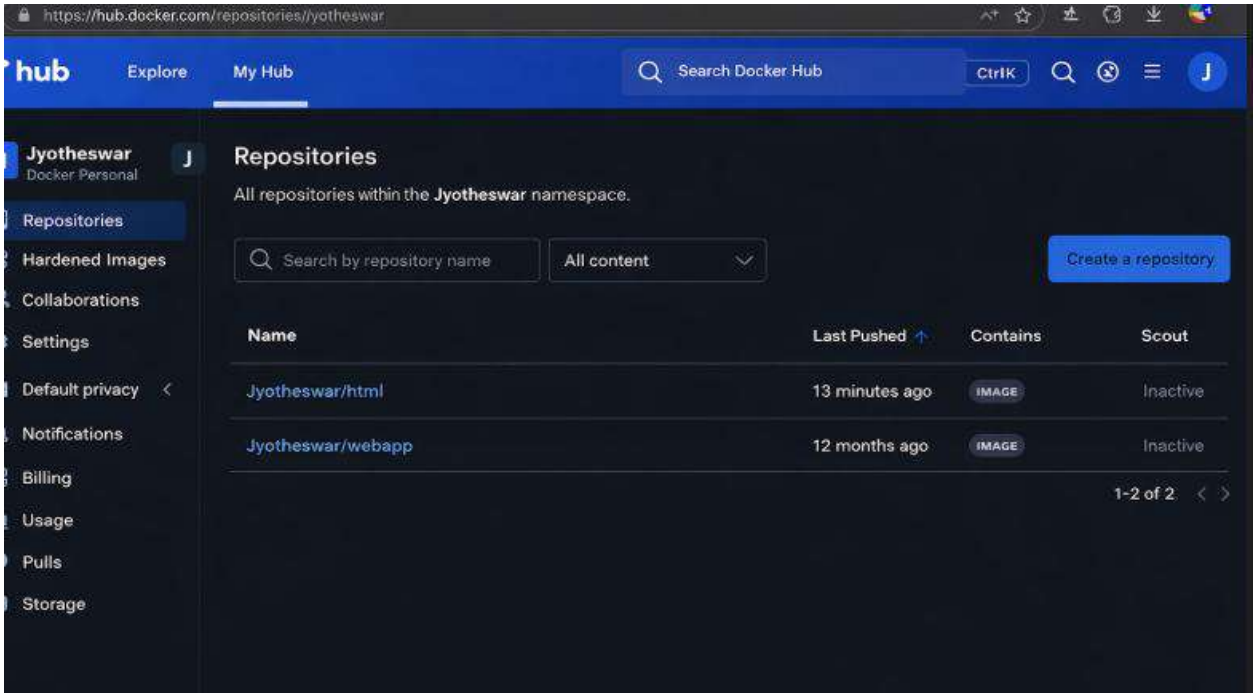
```
pipeline {
   agent any

   environment {
      DOCKER_IMAGE = "druthendra/html-demo"
   }

   stages {
```

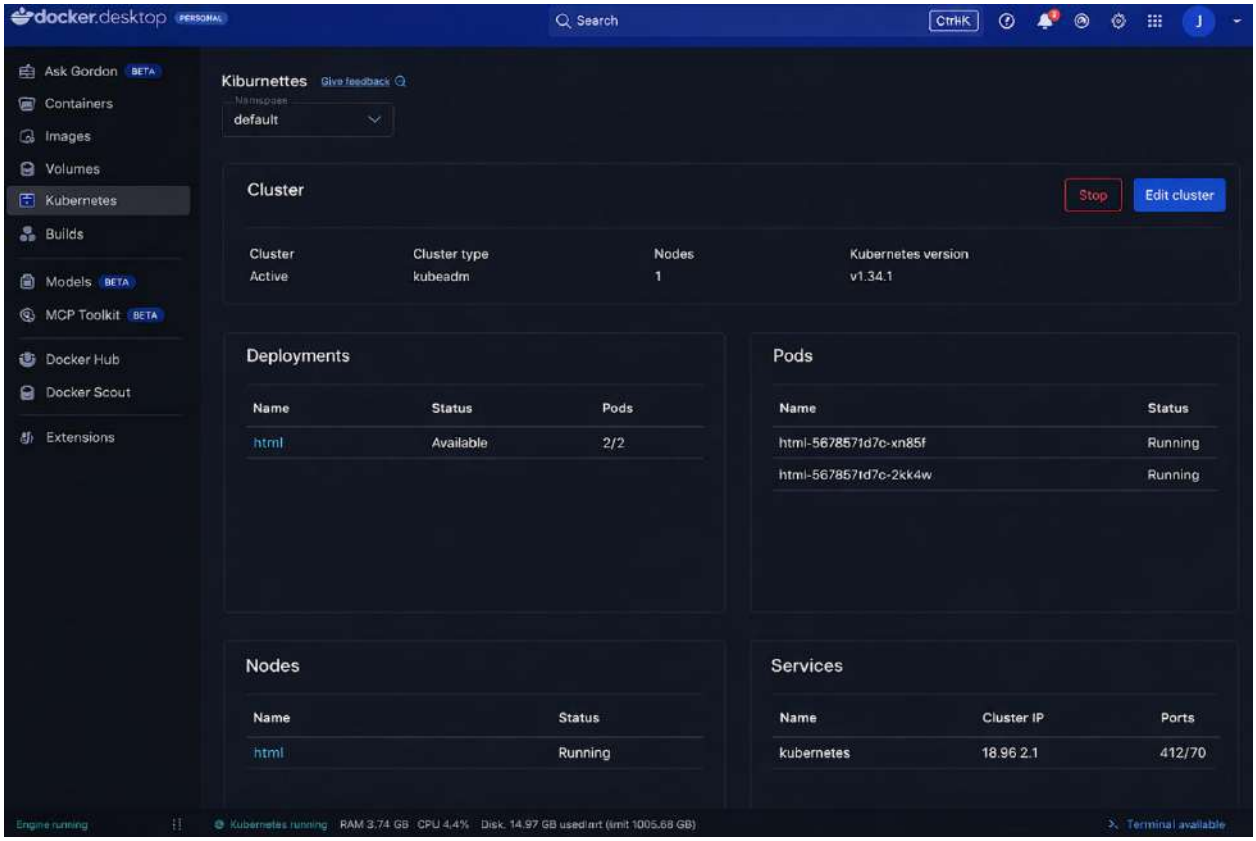
```
        stage('Clone Code') {
            steps {
                git branch: 'main',
                url: 'https://github.com/ethical0101/docker-example.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat 'docker build -t %DOCKER_IMAGE% .'
            }
        }

        stage('Push Image') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'dockerhub',
                    usernameVariable: 'USER',
                    passwordVariable: 'PASS')]) {

                    bat 'docker login -u %USER% -p %PASS%'
                    bat 'docker push %DOCKER_IMAGE%'
                }
            }
        }

        stage('Deploy to Kubernetes') {
            steps {
                withCredentials([file(
                    credentialsId: 'kuberconfig',
                    variable: 'KUBECONFIG')]) {

                    bat '''
                    set KUBECONFIG=%KUBECONFIG%
                    kubectl apply -f deployment.yaml --validate=false
                    '''
                }
            }
        }
    }
}
```

Jenkins
docker-demo
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
Builds
No builds
docker-demo
CI/CD Pipeline using Docker and Kubernetes
Permalinks

Building:-

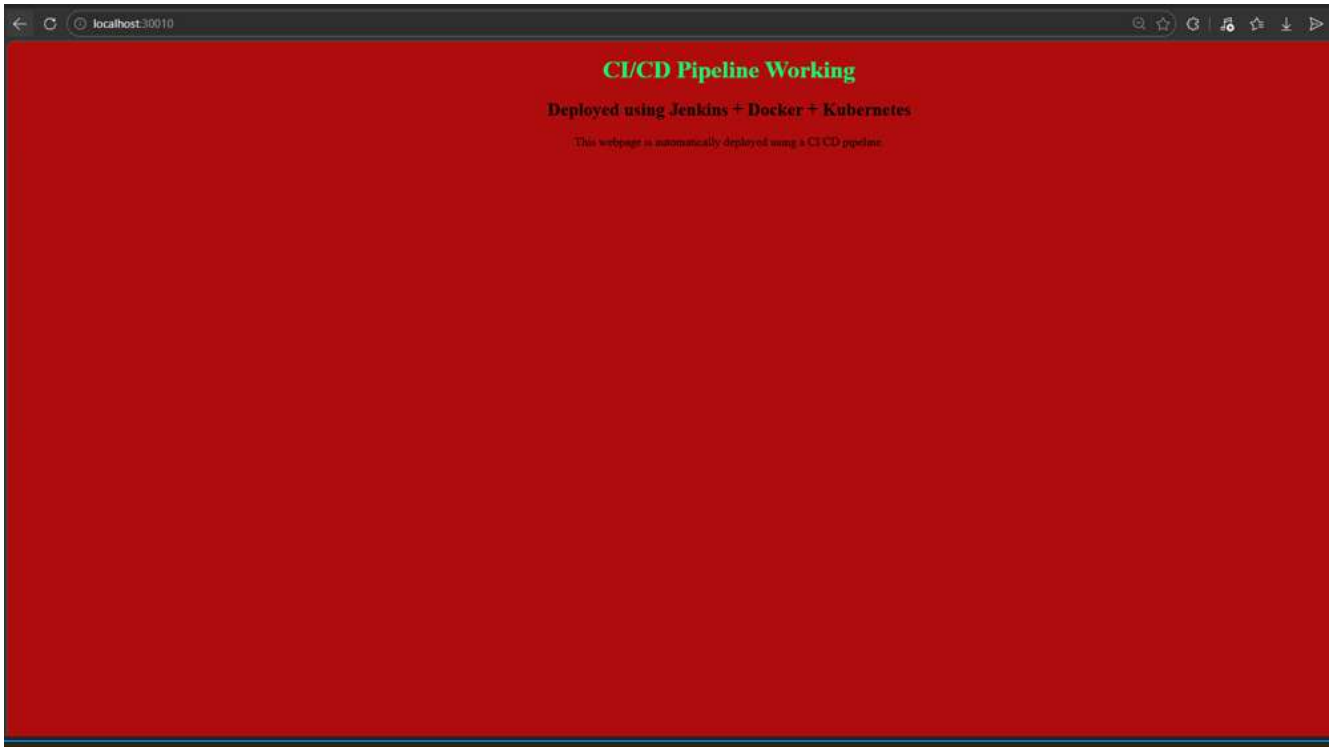

docker-demo
Edit description
CI/CD Pipeline for HTML Demo using Docker and Kubernetes
Stage View
Clone Code
Build Docker Image
Push Image
Deploy to Kubernetes
Average stage times:
(full run time: ~7min 48s)
3s
1min 17s
6min 22s
1s
11
Mar 25
22:56
No Changes
3s
1min 17s
6min 22s
1s
Permalinks
Last build (#1), 11 min ago
Last stable build (#1), 11 min ago
Last successful build (#1), 11 min ago
Last completed build (#1), 11 min ago





Also the HTML page should open in the port mentioned in deployment.yaml file: nodePort: 30010

So check if the HTML file open in http://localhost:30010


localhost:30010
CI/CD Pipeline Working
Deployed using Jenkins + Docker + Kubernetes
This webpage is automatically deployed using a CI/CD pipeline